

USE CASE FLOW – VEHICLE PARKING SYSTEM

UC-01: User Registration / Login

Actor: User, Parking Administrator

Requirement: FR-01

Flow:

1. User or administrator opens the system.
2. User or administrator enters login details.
3. System verifies the credentials.
4. System provides access to the system.

UC-02: Vehicle Registration

Actor: User

Requirement: FR-02

Flow:

1. User logs into the system.
2. User selects vehicle registration.
3. User enters vehicle number, vehicle type, and owner information.
4. System stores the vehicle details.

UC-03: Parking Slot Management

Actor: Parking Administrator

Requirement: FR-03

Flow:

1. Administrator logs into the system.
2. Administrator accesses parking slot management.
3. Administrator adds, updates, or removes parking slots.
4. System updates the parking slot information.

UC-04: Slot Availability

Actor: User

Requirement: FR-04

Flow:

1. User accesses parking availability.
2. System checks the current parking slot status.
3. System identifies available and occupied slots.

4. System displays the slot availability to the user.

UC-05: Vehicle Entry

Actor: User

Requirement: FR-05

Flow:

1. Vehicle arrives at the parking area.
2. System records the vehicle entry time.
3. System checks for an available parking slot.
4. System allocates an available parking slot.
5. System updates the slot status as occupied.

UC-06: Vehicle Exit

Actor: User

Requirement: FR-06

Flow:

1. Vehicle leaves the parking area.
2. System records the vehicle exit time.
3. System identifies the occupied parking slot.
4. System releases the parking slot.
5. System updates the slot status as available.

UC-07: Parking Reservation

Actor: User

Requirement: FR-07

Flow:

1. User accesses parking reservation.
2. System displays available parking slots.
3. User selects an available parking slot.
4. User specifies the reservation period.
5. System records and confirms the reservation.

UC-08: Parking Fee Calculation

Actor: User

Requirement: FR-08

Flow:

1. System obtains the parking duration.
2. System identifies the applicable parking rate.
3. System calculates the parking fee.
4. System displays the calculated fee to the user.

UC-09: Payment Management

Actor: User

Requirement: FR-09

Flow:

1. User views the calculated parking fee.
2. User selects a supported payment method.
3. User makes the payment.
4. System processes the payment.
5. System records the payment status.

UC-10: Parking History

Actor: User

Requirement: FR-10

Flow:

1. User accesses parking history.
2. System retrieves previous parking session records.
3. System displays entry time, exit time, parking slot, and fee.
4. User views the parking history.

UC-11: Notifications

Actor: User

Requirement: FR-11

Flow:

1. System detects an event requiring notification.
2. System generates a notification.
3. System notifies the user about reservation confirmation, parking expiry, or payment status.
4. User receives the notification.

UC-12: Admin Dashboard

Actor: Parking Administrator

Requirement: FR-12

Flow:

1. Administrator logs into the system.
2. Administrator opens the dashboard.
3. System retrieves parking information.
4. System displays parking occupancy, active vehicles, reservations, and revenue.

UC-13: Search Vehicle

Actor: Parking Administrator

Requirement: FR-13

Flow:

1. Administrator accesses vehicle search.
2. Administrator enters the vehicle number or relevant vehicle details.
3. System searches the vehicle records.
4. System displays the matching vehicle information.

UC-14: Reports

Actor: Parking Administrator

Requirement: FR-14

Flow:

1. Administrator accesses the reports section.
2. Administrator selects the required report.
3. System retrieves the relevant parking data.
4. System generates the report.
5. Administrator views the report.